# Continuous Functions

Def'n: Let $X, Y$ be topological spaces. A function $f: X \to Y$ is said to be continuous if for each subset $V$ of $Y$, the set $f^{-1}(V)$ is an open subset of $X$.

If the topology of $Y$ is given by a basis $\mathcal{B}$, then it is sufficient to show that the preimage of every basis element is open, because $V = \bigcup_{\alpha \in J} B_\alpha$, and then $f^{-1}(V) = \bigcup_{\alpha \in J} f^{-1}(B_\alpha)$, which is open if each $f^{-1}(B_\alpha)$ is open.

Theorem: Let $X$ and $Y$ be topological spaces; let $f: X \to Y$. Then

1) $f$ is continuous $\iff$

2) For every $A \subseteq X$, $f(\bar{A}) \subseteq \overline{f(A)}$ $\iff$

3) For every closed set $B \subseteq Y$, $f^{-1}(B)$ is closed in $X$.

Proof: (1 $\Rightarrow$ 2) Assume $f$ is continuous. Let $A \subseteq X$. We must show that $x \in \bar{A} \Rightarrow f(x) \in \overline{f(A)}$. Let $V$ be a neighborhood of $f(x)$. Then $f^{-1}(V)$ is a neighborhood of $x$. Because $x \in \bar{A}$, every neighborhood of $x$ must intersect $A$ in some point $y$. So, $f^{-1}(V) \ni y \Rightarrow V$ intersects $f(A)$ in $f(y)$. So $f(x) \in \overline{f(A)}$.

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(2 $\Rightarrow$ 3) Let $B$ closed in $Y$, $A = f^{-1}(B)$. We wish to show that $A$ is closed in $X$; we show that $\bar{A} \subseteq A$. We know that $f(A) \subseteq B$. Let $x \in \bar{A}$, then $f(x) \in f(\bar{A}) \subseteq \overline{f(A)} \subseteq \bar{B} = B$. which means $f(x) \in f(A) \Rightarrow x \in A$. So $\bar{A} \subseteq A$.

(3 $\Rightarrow$ 1) Let $V$ be open in $Y$. Let $B = Y \setminus V$. Then $B$ is closed in $Y$. So, $f^{-1}(B)$ is closed in $X$ by 3. So, $f^{-1}(V) = f^{-1}(Y \setminus B) = f^{-1}(Y) \setminus f^{-1}(B)$ $= X \setminus f^{-1}(B)$ which is open in $X$. So $f(V)$ is open. $\square$

## Homeomorphisms

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**Def'n:** Let $X$ and $Y$ be topological spaces; let $f: X \to Y$ be a bijection. If $f: X \to Y$ and $f^{-1}: Y \to X$ are continuous, then $f$ is called a homeomorphism.

In other words, $f$ is a homeomorphism when

$$U \text{ open} \iff f(U) \text{ open.}$$

## Topological Imbeddings

**Def'n:** Let $f: X \to Y$ be an injective continuous map, $X, Y$ topological spaces. Let $Z$ be $f(X)$ in the subspace topology in $Y$. Then $f: X \to Z$ defined by restricting the range of $f$ is bijective. If $f$ happens to be a homeomorphism of $X$ with $Z$, then $f$ is called a (topological) imbedding.

## Constructing Continuous functions

**Theorem:** Let $X, Y, Z$ be topological spaces.

a) (Constant) If $f: X \to Y$ maps all of $X$ to $y_0$ in $Y$, then $f$ is continuous.

b) (Inclusion) If $A$ is a subspace of $X$, the inclusion function $j: A \to X$ is continuous.

c) (Composites) If $f: X \to Y$ and $g: Y \to Z$ are continuous, then $g \circ f: X \to Z$ is continuous.

d) (Domain Restriction) If $f: X \to Y$ is continuous and if $A$ is a subspace of $X$, then $f|_A: A \to Y$ is continuous.

e) (Restricting or expanding range) Let $f: X \to Y$ be continuous. If $Z$ is a subspace of $Y$ containing $f(X)$, then $g: X \to Z$ obtained by restricting the range of $f$ is continuous. If $Z$ has $Y$ as a subspace, then $h: X \to Z$ obtained by expanding the range of $f$ is continuous.

f) (Local formulation of Continuity) $f: X \to Y$ is continuous if $X$ can be written as the union of open sets $U_\alpha$ s.t. $f|_{U_\alpha}$ is continuous for each $\alpha$.

g) (Continuity of combined maps) $f: X \to Y$ is a homeomorphism if

$U_\alpha$ s.t. $f|_{U_\alpha}$ is continuous for each $\alpha$.

g) (Continuity at each point) $f: X \to Y$ is continuous if for each $x \in X$, neighborhood $V$ of $f(x)$, there is $U_\alpha$ s.t. $f(U) \subseteq V$.

Proof:

a) Let $f(x) = y_0 \ \forall x \in X$. Let $V$ be open in $Y$. Then if $y_0 \in V$, $f^{-1}(V) = X$ is open. if $y_0 \notin V$, then $f^{-1}(V) = \emptyset$ is open.

b) Let $U$ be open in $X$, then $j^{-1}(U) = U \cap A$ is open in the subspace topology because $U$ open and $A$ open.

c) Let $U$ open in $Z$. Then $g^{-1}(U)$ open in $Y$ and $f^{-1}(g^{-1}(U))$ is open in $X$. $f^{-1}(g^{-1}(U)) = (g \circ f)^{-1}(U)$.

d) $f_{|A}$ is the composite of the inclusion map and $f$ which are both continuous. Follows from c.

e) Let $f: X \to Y$ be continuous. If $f(X) \subseteq Z \subseteq Y$, we show that $g: X \to Z$ obtained from $f$ is continuous. Let $B$ be open in $Z$. Then $B = Z \cap U$ for some $U$ open in $Y$. Because $Z \supseteq f(X)$, $f^{-1}(U) = f^{-1}(U \cap Z) = g^{-1}(B)$. Because $f^{-1}(U)$ is open, $g^{-1}(B)$ is open. Thus, $g$ restricting the range of $f$ is continuous.

Let $h: X \to Z$, where $Y$ is a subspace of $Z$. Then $h$ is the composite of the inclusion map $j: Y \to Z$ and $f: X \to Y$. $h = j \circ f$. Both are continuous, so $h$ is continuous by C.

f) Suppose $X$ can be written as the union of open sets $U_\alpha$ with $f_{|U_\alpha}$ continuous for each $\alpha$. Let $V$ be open in $Y$. Then $f^{-1}(V) \cap U_\alpha = f^{-1}_{|U_\alpha}(V)$. Because each $f_{|U_\alpha}$ is continuous, $V$ is open in $U_\alpha$ and thus open in $X$ because $U_\alpha$ is open in $X$.

$$f^{-1}(V) = \bigcup_\alpha f^{-1}_{|U_\alpha}(V)$$

is a union of open sets and thus open.

g) Let $V$ be open in $Y$, $x \in f^{-1}(V)$. Then $f(x) \in V$.

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g) Let $V$ be open in $Y$, $x \in f^{-1}(V)$. Then $f(x) \in V$. By hypothesis $\exists U_x$ of $x$ s.t. $f(U_x) \subseteq V$. Then $U_x \subseteq f^{-1}(V)$. Thus $f^{-1}(V)$ can be written as a union of open sets each is thus open. □

**Theorem:** The Pasting lemma. Let $X = A \cup B$, with $A, B$ closed in $X$. Let $f: A \to Y$ and $g: B \to Y$ be continuous. If $\forall x \in A \cap B$, $f(x) = g(x)$, then

$$h := \begin{cases} f(x), & x \in A \\ g(x), & x \in B \end{cases}$$

is continuous.

Proof: Let $C$ be closed in $Y$. Then $h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$. Since $f$ is continuous, $f^{-1}(C)$ is closed in $A$. Similarly, $g^{-1}(C)$ is closed in $B$. Because $A, B$ closed in $X$, $f^{-1}(C)$ and $g^{-1}(C)$ are closed in $X$, and hence so is their union. So, $h^{-1}(C)$ is closed in $X$. Thus $h$ is continuous. □

**Theorem:** (maps into products) Let $f: A \to X \times Y$ be given by

$$f(a) = (f_1(a), f_2(a)).$$

Then $f$ is continuous iff the functions

$$f_1: A \to X \quad \text{and} \quad f_2: A \to Y$$

are continuous.

($f_1$ and $f_2$ are called the coordinate functions of $f$)

Proof: let $\pi_1: X \times Y \to X$ and $\pi_2: X \times Y \to Y$ be projections onto the first and second factors. These are continuous, for if $U, V$ open, then $\pi_1^{-1}(U) = U \times Y$, and $\pi_2^{-1}(V) = X \times V$ which are open. Note that $\forall a \in A$,

$$\pi_1 \circ f(a) = f_1(a), \quad \pi_2 \circ f(a) = f_2(a).$$

$\forall a \in A$,

$$\pi_1 \circ f(a) = f_1(a), \quad \pi_2 \circ f(a) = f_2(a).$$

If $f$ is continuous, then by (*) above, the composition of continuous functions is cont. and so $f_1, f_2$ are continuous.

Now suppose that $f_1, f_2$ are continuous. $a \in f^{-1}(U \times V)$ iff $f(a) \in U \times V$, which is equivalent to $f_1(a) \in U$, $f_2(a) \in V$. So,

$$f^{-1}(U \times V) = f_1^{-1}(U) \cap f_2^{-1}(V).$$

The intersection of open sets is open. □

Exercises:

1) Prove that for $f: \mathbb{R} \to \mathbb{R}$, $\varepsilon$-$\delta$ def'n implies the open set definition.

Proof: Suppose $V \subseteq \mathbb{R}$ is open and $f$ is continuous by the $\varepsilon$-$\delta$ def'n. Then $\forall x \in f^{-1}(V)$, $f(x) = y \in V$, $\exists \delta > 0$ s.t. $(y-\delta, y+\delta) \subseteq V$. Then $\exists \varepsilon > 0$ s.t. $f((x-\varepsilon, x+\varepsilon)) \subseteq (y-\delta, y+\delta)$ so $(x-\varepsilon, x+\varepsilon) \subseteq f^{-1}(V)$. □

2) Suppose that $f: X \to Y$ is continuous. If $x$ is a limit point of $A \subseteq X$, is $f(x)$ a limit point of $f(A)$?

Proof: $f|_A$ is continuous by Thm 7.2. $x$ is a limit point of $A \Rightarrow \forall U_x$, $U_x \cap (A \setminus \{x\}) \neq \emptyset$. Suppose $x$ is a limit point of $A$. We wish to show that for $V$ neighborhood of $f(x)$, $V \cap (f(A) \setminus \{f(x)\}) \neq \emptyset$. Let $V$ be a neighborhood of $f(x)$. Then $f^{-1}(V)$ is a neighborhood of $x$, so $f^{-1}(V) \cap (A \setminus \{x\}) \ni z$. So $f(z) \in V \cap f(A \setminus \{x\}) \subseteq V \cap f(A)$. So $f(x) \in \overline{f(A)}$. Suppose for all $y \in A$, $f(y) = f(x)$. Then $f(z) \in V \cap \{f(x)\}$ only intersects at $f(x)$ and is thus not a limit point. □

4) Given $x_0 \in X$, $y_0 \in Y$, show that $f: X \to X \times Y$ and $g: Y \to X \times Y$ by $f(x) = x \times y_0$ and $g(y) = x_0 \times y$ are imbeddings

$g: Y \to X \times Y$ by $f(x) = x \times y_0$ and $g(y) = x_0 \times y$ are imbeddings.

Proof: We wish to show that $f: X \to X \times \{y_0\}$ is bijective and continuous.

(Injective).

Suppose $f(x) = f(y)$.

$(x, y_0) = (y, y_0) \Rightarrow x = y$. ✓

(Surjective).

Suppose $f(x) = (y, y_0)$, $y \in X$.

Then $x = y$ gives $f(y) = (y, y_0)$. ✓

Continuous: $X \times \{y_0\}$ is open in the subspace topology. Let $V$ be open in $X \times \{y_0\}$. Then $V = A \times \{y_0\}$ where $A$ is open in $X$.

$f^{-1}(V) = A$ which is open. So $f$ is continuous.

Now we show that $f^{-1}$ is continuous. Let $V$ be open in $X$. In the Product topology, $A \times B$ is open iff $A$ is open and $B$ is open. $\{y_0\}$ is open in the subspace topology $\{y_0\} \subseteq Y$, $V$ is open in $X$, so $V \times \{y_0\}$ is open in $X \times \{y_0\}$.

Thus $f$ is continuous.

Therefore $f$ is an embedding.

(or) Let $V \subseteq X$ be open. We must show that $f(V) = V \times \{y_0\}$ is open. This is open because $V \times \{y_0\} = V \times Y \cap X \times \{y_0\}$, and $V \times Y$ is open in $X \times Y$ because $V$ open in $X$ and $Y$ open in $Y$. So $f(V)$ is open when $V$ is open.

Thus $f^{-1}$ and $f$ are continuous.

□

5)

(a) Let $X$ and $Y$ be ordered sets under the order topology. Show that if $f: X \to Y$ is order preserving, then it is a homeomorphism.

then it is a homeomorphism.

Proof: $f$ order preserving means if $a \le x \le b$, then $f(a) \le f(x) \le f(b)$, and $a < x < b$ then $f(a) < f(x) < f(b)$.

Let $V$ be a neighborhood of $f(x)$ in $Y$. Then $V$ is open in $Y$. Any open set in $Y$ is of the form $\bigcup (a,b)$ s.t. $a < b$. Because $f$ is order preserving, each $a, b \to f(a), f(b)$, and $f(a) < f(b)$. So $f^{-1}(a,b) \to (f^{-1}(a), f^{-1}(b))$ $f(x)$ must be in at least one interval, so $x \in f^{-1}(V)$. So, $f^{-1}(V)$ is a union of open intervals in $X$ and thus open. □

b) Let $n \in \mathbb{Z}^+$. Suppose $\forall x \in \mathbb{R}^+ \cup \{0\}$, there is a unique $b \in \mathbb{R}^+ \cup \{0\}$ s.t. $b^n = x$. Denote $b$ by $\sqrt[n]{x}$. Show that $g: \overline{\mathbb{R}^+} \to \overline{\mathbb{R}^+}$ defined by $g: x \mapsto \sqrt[n]{x}$ is continuous.

Proof: We show that $g$ is order preserving. Suppose $a < b \in \overline{\mathbb{R}^+}$. $g(a) = \sqrt[n]{a}$, $g(b) = \sqrt[n]{b}$. By exercise, $g(a) < g(b)$. Because $g$ is order preserving, by (a) it is a homeomorphism and thus continuous. □

6) Show that the subspace $(a,b)$ of $\mathbb{R}$ is homeomorphic with $(0,1)$, and $[a,b]$ is homeomorphic with $[0,1]$.

Proof: Let $f: (0,1) \to (a,b)$ be defined by $f: x \mapsto a + (b-a)x$. Let $z \in (a,b)$. $a + (b-a)x = z \Rightarrow (b-a)x = z - a$. So $x = (z-a)(b-a)^{-1}$. $z - a < b - a$, so $x \in (0,1)$. Thus $f$ is surjective. Suppose $f(x) = f(y)$. Then $a + (b-a)x = a + (b-a)y \Rightarrow x = y$. So $f$ is injective. Thus, $f$ is bijective. We now show that $f$ is order preserving. Suppose $x \le y$. Then $f(x) = a + (b-a)x$, $f(y) = a + (b-a)y$. $x \le y \Rightarrow (b-a)x \le (b-a)y$. and so $a + (b-a)x \le a + (b-a)y$, is $f(x) \le f(y)$. So, $f$ is order preserving and bijective. $f$ is thus a homeomorphism from $(0,1) \to (a,b)$. The case for $f: [0,1] \to [a,b]$ is identical. □

9) Let $Y$ be an ordered set in the order topology. Let $f, g: X \to Y$ be continuous.

a) Show that $\{x \mid f(x) \leq g(x)\}$ is closed in $X$.

Proof: Notice that $\{x \mid f(x) \leq g(x)\} = f^{-1}\{f(x) \mid f(x) \leq g(x)\} \cup g^{-1}\{g(x) \mid g(x) \geq f(x)\}$. Furthermore, because both $f, g$ map $X$ to $Y$, these sets are identical. So, we show that $\{f(x) \mid f(x) \leq g(x)\}$ is closed in $Y$. This set is equivalent to $(-\infty, g(x)] \subseteq Y$. $Y \setminus (-\infty, g(x)] = (g(x), \infty)$ is open, so $\{f(x) \mid f(x) \leq g(x)\}$ is closed in $Y$, and thus by continuity its preimage is closed in $X$.

b) Let $h: X \to Y$ be the function $h(x) = \min\{f(x), g(x)\}$. Show that $h$ is continuous.

Proof: Consider the sets $A = \{x \mid f(x) \leq g(x)\}$, $B = \{x \mid g(x) \leq f(x)\}$. $A \cup B = X$, because $f$ and $g$ are both defined entirely on $X$. Because $f, g$ continues on $X$, they are continuous on $A, B \subseteq X$. Consider $f|_A, g|_B$. Suppose $x \in A \cap B$. Then $f(x) \leq g(x) \leq f(x) \Rightarrow f(x) = g(x)$.

$h = \begin{cases} f|_A, & x \in A \\ g|_B, & x \in B \end{cases}$ is equivalent to $h$ defined above and is thus continuous from $X$ to $Y$ by the pasting lemma. □

10) Let $\{A_\alpha\}$ be a collection of subsets of $X$. Let $X = \bigcup_\alpha A_\alpha$. Let $f: X \to Y$. Suppose that $f|_{A_\alpha}$ is continuous for each

Let $f: X \to Y$. Suppose that $f|_{A_\alpha}$ is continuous for each $\alpha$.

a) Show that if $\{A_\alpha\}$ is finite and each $A_\alpha$ is closed, then $f$ is continuous.

Proof: Suppose $|\{A_\alpha\}| = 2$, $A_1, A_2$ closed. We can apply the pasting lemma, for clearly $f|_{A_1}(x) = f|_{A_2}(x)$ for all $x \in A_1 \cap A_2$. Thus $h(x) = f|_{A_1}(x)$ if $x \in A_1$, $h(x) = f|_{A_2}(x)$ if $x \in A_2$ is just $f(x)$. Thus $f$ is continuous. Now suppose $\{A_\alpha\}$ is finite and each $A_\alpha$ is closed, $X = \bigcup_\alpha A_\alpha$. Then as we have shown, this is $A_1 \cup A_2 \cup (\bigcup_{\alpha \in \{1,2,3\}} A_\alpha)$. $f|_{A_1 \cup A_2}$ is continuous. You can now induct to get $(A_1 \cup A_2) \cup A_3 \cup (\bigcup_{\alpha \in \{1,2,3\}} A_\alpha)$ and have $f|_{A_1 \cup A_2 \cup A_3}$ continuous. Inducting gives

$f|_{\bigcup A_\alpha}$ continuous, because $X = \bigcup_\alpha A_\alpha$,

we have that $f|_X = f$ is continuous. □

b) Find an example where $\{A_\alpha\}$ is countable and each $A_\alpha$ is closed but $f$ is not continuous.